

Systems of Linear Differential Equations

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from converting a higher-order ODE into a first-order system, through the eigenvalue method and the full decoupling $A = PDP^{-1}$ with P , P^{-1} , and D computed explicitly, into the matrix exponential $e^{At} = Pe^{Dt}P^{-1}$, the fundamental matrix and its link $e^{At} = \Psi(t)\Psi(0)^{-1}$, and a complex-eigenvalue capstone that produces a spiraling real solution. Solutions follow in Part III.

1. **(To a system)** Convert $y'' + 5y' + 6y = 0$ into a first-order system $\mathbf{x}' = A\mathbf{x}$, and state the companion matrix A .
2. **(Eigenvalues)** For $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$, compute $\det(A - \lambda I)$ and find both eigenvalues.
3. **(Eigenvectors and P)** For that same A , find an eigenvector for each eigenvalue and assemble the modal matrix $P = [\mathbf{v}_1 \mid \mathbf{v}_2]$.
4. **(Inverse)** Compute P^{-1} for $P = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$, showing $\det P$ explicitly.
5. **(Verify decoupling)** Confirm $P^{-1}AP = D$ for the A , P , P^{-1} above, and state the decoupled system $\mathbf{y}' = D\mathbf{y}$.
6. **(General solution)** Write the general solution $\mathbf{x}(t)$ of $\mathbf{x}' = A\mathbf{x}$ from the eigenpairs $\lambda_1 = 4$, $\mathbf{v}_1 = (2, 3)$ and $\lambda_2 = -1$, $\mathbf{v}_2 = (1, -1)$.
7. **(Matrix exponential)** Using $A = PDP^{-1}$ with $D = \text{diag}(4, -1)$, build $e^{At} = Pe^{Dt}P^{-1}$ in closed form.
8. **(Fundamental matrix)** Write a fundamental matrix $\Psi(t)$ for this system and compute its Wronskian $\det \Psi(t)$.
9. **(Normalization)** Explain why $e^{At} = \Psi(t)\Psi(0)^{-1}$, and identify $\Psi(0)$ for the Ψ above.
10. **(Complex capstone)** Solve $\mathbf{x}' = A\mathbf{x}$ for $A = \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$, giving the real general solution and the spiral direction.

Part III — Complete Solutions

Solution 1. Let $x_1 = y$, $x_2 = y'$, so $x'_1 = x_2$ and $x'_2 = y'' = -6y - 5y' = -6x_1 - 5x_2$:

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -6 & -5 \end{pmatrix}.$$

The coefficients enter the bottom row with flipped signs.

Solution 2. Expand the determinant:

$$\det(A - \lambda I) = (1 - \lambda)(2 - \lambda) - (2)(3) = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1).$$

So $\lambda_1 = 4$ and $\lambda_2 = -1$.

Solution 3. Solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue:

$$\lambda = 4: \begin{pmatrix} -3 & 2 \\ 3 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_1 = (2, 3);$$

$$\lambda = -1: \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_2 = (1, -1).$$

Hence $P = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$.

Solution 4. The determinant and inverse are

$$\det P = (2)(-1) - (1)(3) = -5, \quad P^{-1} = \frac{1}{-5} \begin{pmatrix} -1 & -1 \\ -3 & 2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ 3 & -2 \end{pmatrix}.$$

Solution 5. Multiply in two steps, AP then $P^{-1}(AP)$:

$$AP = \begin{pmatrix} 8 & -1 \\ 12 & 1 \end{pmatrix}, \quad P^{-1}AP = \frac{1}{5} \begin{pmatrix} 20 & 0 \\ 0 & -5 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} = D.$$

The decoupled system is $y'_1 = 4y_1$, $y'_2 = -y_2$.

Solution 6. Each eigenpair contributes one exponential mode:

$$\mathbf{x}(t) = c_1 e^{4t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

With $\lambda_1 = 4 > 0 > \lambda_2 = -1$ the origin is a saddle.

Solution 7. Exponentiate the diagonal and reassemble $e^{At} = P e^{Dt} P^{-1}$:

$$P e^{Dt} = \begin{pmatrix} 2e^{4t} & e^{-t} \\ 3e^{4t} & -e^{-t} \end{pmatrix}, \quad e^{At} = \frac{1}{5} \begin{pmatrix} 2e^{4t} + 3e^{-t} & 2e^{4t} - 2e^{-t} \\ 3e^{4t} - 3e^{-t} & 3e^{4t} + 2e^{-t} \end{pmatrix}.$$

At $t = 0$ this is I , as it must be.

Solution 8. Stack the two eigensolutions as columns:

$$\Psi(t) = \begin{pmatrix} 2e^{4t} & e^{-t} \\ 3e^{4t} & -e^{-t} \end{pmatrix}, \quad \det \Psi = -2e^{3t} - 3e^{3t} = -5e^{3t} \neq 0.$$

The nonzero Wronskian confirms the columns are independent.

Solution 9. Both $\Psi(t)$ and e^{At} solve $X' = AX$; they agree once normalized to the same value at $t = 0$. Since $e^{A \cdot 0} = I$,

$$e^{At} = \Psi(t) \Psi(0)^{-1}, \quad \Psi(0) = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix} = P.$$

Right-multiplying by $\Psi(0)^{-1}$ forces the value I at $t = 0$.

Solution 10. Here $\det(A - \lambda I) = \lambda^2 - 4\lambda + 13 = 0$ gives $\lambda = 2 \pm 3i$. With $\mathbf{v} = (1, -i)$ for $\lambda = 2 + 3i$, Euler's formula yields

$$\mathbf{x}(t) = e^{2t} \left[c_1 \begin{pmatrix} \cos 3t \\ \sin 3t \end{pmatrix} + c_2 \begin{pmatrix} \sin 3t \\ -\cos 3t \end{pmatrix} \right].$$

Since $\alpha = 2 > 0$, every trajectory spirals outward.

Unit Key Takeaway

One idea threads the whole unit: a linear system is easiest in its own eigenbasis. Recast higher-order ODEs as $\mathbf{x}' = A\mathbf{x}$, then **diagonalize** $A = PDP^{-1}$ so the modal substitution $\mathbf{y} = P^{-1}\mathbf{x}$ **decouples** the system. The **matrix exponential** $e^{At} = Pe^{Dt}P^{-1}$ packages every mode into one operator and generalizes scalar e^{at} ; the **fundamental matrix** collects the same solutions and reproduces e^{At} at $t = 0$. Complex eigenvalues simply trade pure exponentials for spirals.